



Data Article

A fine-grained labeled dataset for textual sentiment analysis in technical education

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ABSTRACT

This paper presents a meticulously curated dataset tailored for textual sentiment analysis within the realm of technical education, falling under the domain of Natural Language Processing and Pattern Recognition. The dataset, crafted in collaboration with the All India Council for Technical Education (AICTE), encompasses over 14,000 records manually entered by representatives from technical institutes across India over the course of one year. The data, hosted on AICTE's in-house servers, has been categorized into seven distinct labels, including Appreciation, Complaint, Support, Suggestion, among others. Through a detailed data collection process facilitated by an online application, this dataset serves as a cornerstone for sentiment analysis within the domain of technical education. Notably, it is the first publicly available dataset of its kind, providing a rich resource for evaluating existing models and fostering the development of novel ones. The dataset, consisting of 14,272 records, is further enhanced with classification into 10 distinct modules, offering a nuanced understanding across various aspects of technical education. The paper outlines the experimental design, materials, and methods employed in the data collection process, along with its limitations and ethical considerations. Additionally, the paper acknowledges the contributions of the All India Council for Technical Education in facilitating the data collection process. This dataset holds significant value in advancing research and

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applications in sentiment analysis and related fields within the domain of technical education.

Approximately 10,000 technical institutions in India operate under the jurisdiction of the All India Council for Technical Education (AICTE). To gather comprehensive data, an extensive online application with ten modules has been devised and distributed to all 10,000 institutions. Over the course of a year, labeled data has been systematically collected from these technical institutions, categorizable into seven distinct types such as Appreciation, Complaint, Support, Suggestion, and more. This rich dataset holds significant potential for applications in deep learning, including sentiment analysis and classification problems. The data, meticulously entered by representatives from these technical institutions, is stored in a highly accurate and systematic process. This dataset, characterized by its precision and reliability, stands as an excellent resource for both training and testing purposes in various deep-learning models. Its suitability extends to applications such as sentiment analysis, where the quality and authenticity of the data are crucial for robust model development.

As best of our knowledge this is first kind of dataset in domain of technical education with 14,000 + samples. Very few Multiclass Multi-labeled dataset are available therefore this dataset is very much useful in applications of Natural Language Processing like Sentiment Analysis etc.

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Specifications Table

Subject	Computer Science
Specific subject area	Textual sentiment analysis that comes under Natural Language Processing and Pattern Recognition
Data format	Raw, Analyzed, Filtered
Type of data	Table
Data collection	A detailed data collection process through an online application has been developed and circulated among all 10,000 technical institutions of India to collect the data. Multiclass Multi-labeled data from these technical institutions have been collected in a year long duration. This data can be classified into seven different categories i.e. Appreciation, Complaint, Support, Suggestion etc. To normalize the data, all data which are either very small or very big in size are removed.
Data source location	All the data are collected with help of the All India Council for Technical Education (AICTE-28.535937, 77.154246) and stored in its in-house database servers. AICTE has its headquarters at Nelson Mandela Road, Vasant Kunj, New Delhi, INDIA 110,070.
Data accessibility	Repository name: Harvard Dataverse Data identification number: 10.7910/DVN/EITGBB Direct URL to data: https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/EITGBB Instructions for accessing these data: <i>There are three files available:</i> one in PKL format and two in Microsoft Excel format. The PKL file enables direct usage of the data within Python code. The first Excel file encompasses the complete dataset, while the second Excel file comprises tables containing diverse label categories and metadata.

1. Value of the Data

- Representatives from technical institutes in India manually entered over 14,000 records within a year. These records were initially marked with seven labels. To the best of our knowledge, this will be the first publicly available dataset about technical education that can be used for sentiment analysis.
- This fine-grained labeled dataset, consisting of 14,272 records, is marked with three classes of labels, including 7 types of labels for categories, 10 kinds of labels for modules, and 42 types of labels for nature-wise labeled data. These diverse labels contribute to the dataset's richness, enabling models to discern and learn relevant features from the input data with precision and specificity.
- This labeled dataset will play a crucial role in machine learning, this dataset would be used in sentiment analysis and other machine learning models. This labeled dataset is a cornerstone in supervised learning, enabling model training, evaluation, and improvement. It empowers machine learning models to make accurate predictions and generalize well to new, unseen data.
- This dataset is being utilized in the BiERU model. The code is publicly available on GitHub (<https://github.com/manoj Singh07/BiERU>) and will be published soon. The data is also provided in PKL file format (<https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/EITGBB>), making it compatible for use in any Python or PyTorch-based project. Dataset is publicly available with Creative Commons Attribution 4.0 International License (CC BY 4.0).

2. Background

In the realm of the domain of sentiment analysis, the demand for a robust dataset is essential for assessing existing models and facilitating the development of new ones. To meet this requirement, a proposed dataset has been meticulously crafted, incorporating manual entries comprising more than 14,000 records. This dataset, created in collaboration with the All India Council for Technical Education, is enriched with seven distinct labels, offering a comprehensive and diverse set of examples for evaluating and implementing both existing and novel sentiment analysis models. Data in .csv file is very useful for researchers [1].

The primary motivation for collecting this dataset, which focuses on fine-grained sentiment analysis in the context of technical education, stems from the scarcity of datasets tailored to this specific domain. Most existing sentiment analysis datasets are either too broad or centered around general topics, creating a significant gap in more specialized areas like technical education. This dataset helps to address the research gaps like Domain-Specific Focus, Fine-Grained Sentiment Classification, and Understanding Contextual Sentiments in Education etc.

3. Protocol for Creating and Collecting Data

The following steps were used while collecting data and to prepare a useful dataset for sentiment analysis.

1. Data Acquisition: Different web forms for user registration and data collection were created using *PHP* and Oracle Siebel. The links to these web forms, along with detailed instructions, were circulated multiple times via email and social media among all technical institutions of India.
2. The authors created web forms, and a few sample forms are displayed below.
3. Data Preprocessing: Manually clean the data by removed unnecessary symbols and content written in languages other than English. Normalize the data by removing very large and very small feedback/ survey received from institutions.

Table 1
Label wise count of record in dataset.

Sr No.	Category of input	Count of input
1	Appreciation	770
2	Complaint	585
3	Enquiry	1244
4	Grievance	6844
5	Required Support	2321
6	Suggestion	1056
7	Willing to Support	1452
Grand Total		14,272

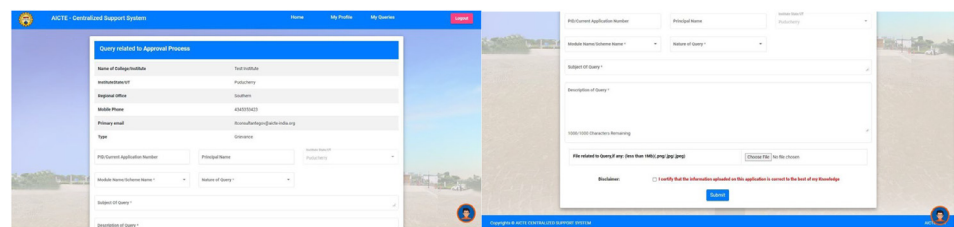


Fig. 1. Samples of the webform.

4. During the data preparation process, all identifying information related to any institute or faculty was removed to ensure the confidentiality of such details.
5. Data Annotation: Manually annotated the collected data in multi-class classification based on various labels.
6. Data Storage and Management: Prepare the metadata and store dataset with meta data in different CSV files. Prepare Pkl file for ready use dataset in python code.

4. Data Description

This article outlines a dataset consisting of feedback from technical institutes across India. Over 14,000 records were manually entered by representatives from these institutes, spanning the entire country. While generative AI is a powerful tool for creating data, real human-entered data is vital for ensuring accuracy, authenticity, and ethical integrity in AI models and applications. All the inputs were in English language. From an English linguistic perspective, the composition of datasets in sentiment analysis or any linguistic study involves several key considerations, such as language variety, lexical diversity, syntax and grammar, and context and pragmatics. These factors are essential to ensure that the data is both representative and comprehensible to a broad audience. The comprehensive dataset was compiled over the course of one year. To eliminate missing values, any records containing them were entirely removed. The records are categorized into seven distinct labels for a more nuanced analysis as shown in [Table 1](#) and [Fig. 1](#).

This data set is created to analyses the sentiments of the feedback submitted by faculty members and other staff members of AICTE stakeholders or technical institutions across India. Representatives from various higher technical institutions across India manually entered these records.

Fine-grained labelled data set is used because Fine-grained sentiment analysis is a more detailed and nuanced approach to sentiment analysis that goes beyond broad categories like “positive,” “negative,” or “neutral.” Instead, it focuses on identifying and classifying more specific sentiments like “appreciation”, “complaint”, “support”, or “suggestion” etc. This level of granularity can provide more actionable insights, especially in fields like market research, customer feedback analysis, and social media monitoring, where understanding the nuanced sentiments of users can be crucial for decision-making.

Table 2
Module wise count of record in dataset.

Sr No.	Module name	Count of Module data
1	Course	1998
2	Deficiency	661
3	DSC Utility	972
4	Faculty	1628
5	Login Issue	586
6	Other	3342
7	Payment	1946
8	Questionnaire	681
9	Report	1862
10	Submit Application	596
Grand Total		14,272

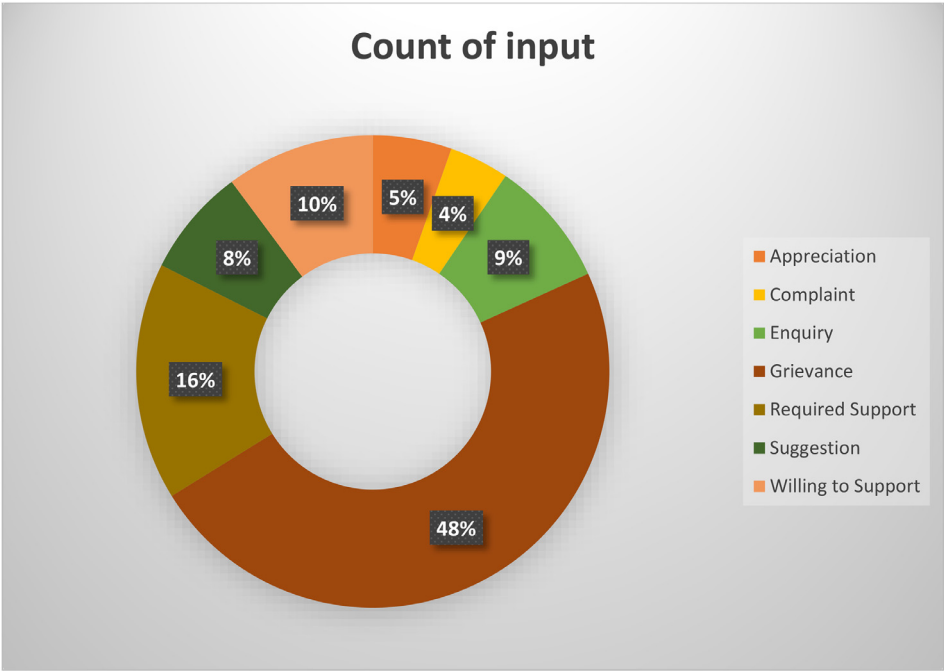


Fig. 2. Doughnut pie chart for labelled data.

These 14,272 records have been further enhanced by classification into 10 distinct modules (complete list is given in Table 2) i.e. Course, Deficiency, Digital Signature Certificate (DSC) Utility, etc. This additional labelling provides a more detailed and nuanced categorization, allowing for a comprehensive understanding and analysis of the dataset across various aspects. Pie chart for module wise data is given in Figs. 2 and 3

Following 42 Nature wise data labels are also available for use (list given in Table 3)

All this information in CSV file format with character count based statistics is also available on public domain on Harvard Dataverse.

Complete 14,272 records are available in one single excel file with following seven columns:

1. Module Scheme Name: Input is collect on based on different AICTE work which can be divided in to 10 different modules

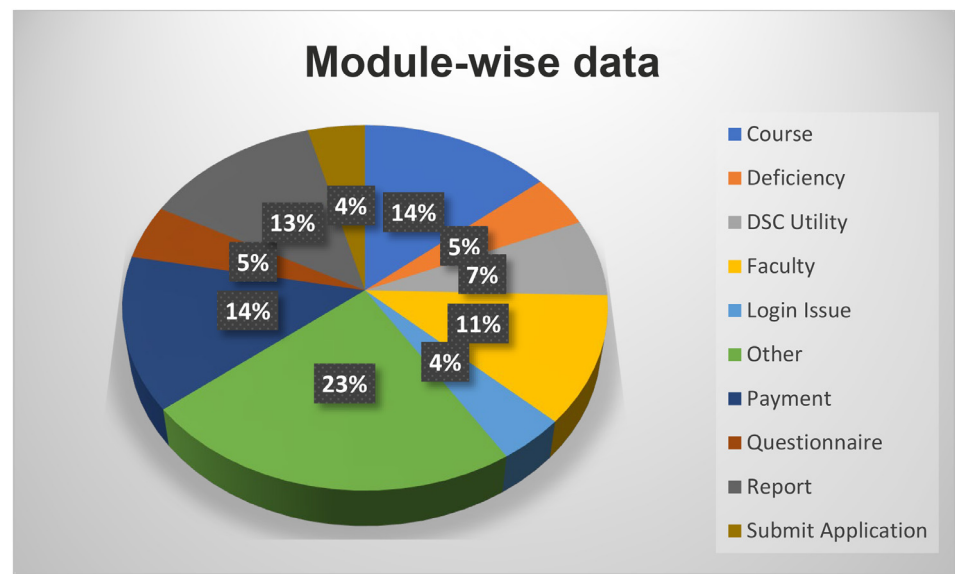


Fig. 3. Module-wise data.

- Nature: Individual institute can select any module out of 10 modules of first column and then select nature of work out of 42different nature of work possible in these 10 modules for submitting their responses.
- Subject: Subject was given be individual institute on based of their response. It is basically heading of their response or feedback.
- Description: Description is actual response or feedback given by the institutions.
- Character Count: Count number of characters of given description.
- Type: Type is main category of feedback or response chosen by institution out of given only seven categories.
- Created On: It gives the actual date and time in Indian Standard Time (IST) when an institution recorded their feedback online.

5. Experimental Design, Materials and Methods

The field of technical education was chosen as the domain for acquiring authentic data from individuals. Consequently, a web form was specifically crafted to serve as a platform for data collection, integrated into the portal of the All India Council for Technical Education (AICTE). The web form was structured to collect responses from various stakeholders, such as students, faculty, and administrators, about their experiences and perceptions related to technical education. The link to this web form was disseminated to all AICTE-approved technical institutions throughout India. These institutions were invited to contribute valuable data across various modules, categories, and labels by selecting their desired options and entering the corresponding information. Representatives from these institutions, who may be students, faculty, and administrators, provided feedback, comments, or questions and actively participated in the data collection process.. Impressively, input from more than 10,000 AICTE-approved institutes has been received successfully. The web server housing the data is hosted on the in-house portal of AICTE.

The web form consisted of ten primary modules, as shown in Table 2, and these modules consisted of six labels, as provided in Table 1. This data collection web form was crafted using PHP and Oracle Siebel, facilitating the gathering of information, which was subsequently

Table 3

Nature of feedback wise count of record in dataset.

Sr No.	Nature of Feedback	Count of Nature wise data
1	Any other tab	270
2	Any Technical queries	2
3	Claim for Excess Payment made	16
4	complaint against institute	3
5	complaint related	41
6	Deficiency related	11
7	Deficiency related Query	475
8	DSC (Digital Signature Certificate) Connection Issue	143
9	DSC Issue During Submit	23
10	DSC Registration Issue	557
11	Error in creating new application	28
12	Faculty record related	242
13	Faculty related Query	1190
14	Forgot Password not received	19
15	In contact tab	13
16	Invalid User Name and Password	35
17	Issue in Course Intake	506
18	Issue in Course/program name	348
19	Issue in Downloading of EoA (Extension of Approval)	980
20	Issue in Downloading of LOA (Letter of Approval)	28
21	Issue in DSC Registration	8
22	Issue in Security Deposit	82
23	Issue in uploading e-Affidavit	61
24	Issue related to NRI (Non Resident Indian)/OCI (Overseas Citizen of India)/INTEGRATED Course	12
25	New User Id and password not received	61
26	Others	5372
27	PAN related	34
28	Payment	1
29	Payment Updation Issue	756
30	Payment/TER (Technical Education Regulatory) Charges related Query	895
31	Document Signing Issue	129
32	policy related	60
33	Policy related Query	3
34	Portal related update	1
35	Program/Course related query	776
36	Quality Education Mandate related Query	7
37	Questionnaire related Query	255
38	Report Generation Issue	495
39	salary related	22
40	SC (Scrutiny Committee)/EVC (Expert Visit Committee)/SAC (Standing Appellate Committee) /SHC (Standing Hearing Committee) section	48
41	Signed (PDF file format) Verification Issue	40
42	Unable to submit the Application	224
Grand Total		14,272

stored in an Oracle database. To enhance accessibility, the Oracle datasets were converted into Microsoft Excel format. This Excel file is made publicly available in the public domain, providing a transparent avenue for access to the dataset [2].

Additionally, for direct utilization in Python-based applications [3], a subset of the data has been converted into a PKL file, streamlining its compatibility with Python environments. This diversified format availability ensures flexibility in accessing and utilizing the dataset across different platforms and applications [4,5].

During this study, the sentiments expressed in the feedback received from institutional representatives were classified into seven different sentiment categories given in Table 1. The initial

dataset consisted of 14,272 responses collected from AICTE-approved technical institutes. However, on classifying the responses, a percentage-wise overview is given in Fig. 2.

As given in Table 1, out of the seven categories of input data, the minimum number of responses is 585 and the maximum number of responses is 6844. Getting a significant number of data on each category was an arduous and most difficult task with this data set, which needed multiple iterations of communication and requests to submit data on various categories. In case additional data is needed to train a model, any of the available Textual augmentation techniques can be employed viz Synonym Replacement, Random Insertion, Random Deletion, Back Translation, and Sentence Shuffling

6. Data Storage and Transformation Process

In this study, data was collected and initially stored in an Oracle database. This choice was made to leverage Oracle's robust data management capabilities, ensuring data integrity, security, and efficient querying. After the data collection phase, the data have been transformed into two additional formats: Excel and PKL (Python Pickle). The rationale behind this multi-format storage strategy, along with the transformation process, is outlined below:

6.1. Rationale for format choices

Excel Format: The conversion of the data into Excel format was primarily for ease of access and usability. Excel is a widely used tool for data analysis and visualization, making it easier for researchers and stakeholders to explore the dataset, perform preliminary analyses, and generate reports without requiring specialized database knowledge.

PKL Format: The selection of PKL (Pickle) format was driven by its compatibility with Python-based applications and its ability to serialize complex data structures, such as lists and dictionaries. Using PKL allows for efficient loading and manipulation of data within Python, facilitating advanced data processing, analysis, and machine learning tasks.

Transformation Process: The transformation of data from the Oracle database to the Excel and PKL formats involved following three steps:

1. **Data Extraction:** SQL queries have been utilized to extract the relevant datasets from the Oracle database. This step ensured selective retrieval of the necessary data points while maintaining data integrity.
2. **Data Cleaning and Pre-processing:** In this step inconsistencies, missing values, or formatting issues have been solved. This step was crucial for ensuring the quality of the dataset.
3. **Conversion to Excel:** The cleaned data was then exported to Excel format using Python's pandas library. This allowed for seamless export and preservation of the dataset structure, including any relevant headers and metadata.
4. **Conversion to PKL:** Following the Excel conversion, the cleaned dataset was serialized into PKL format using the pickle, pandas and csv modules in Python. This process involves converting Python objects into a byte stream, allowing for efficient storage and retrieval. The code snippet is given below:

```
df = pd.read_csv('/content/drive/MyDrive/GitHub/BiERU/ACSS.csv',
encoding='ISO-8859-1', on_bad_lines='skip')
df.to_pickle('/content/drive/MyDrive/GitHub/BiERU/ACSS.pkl')
```

Verification: After conversion, verification checks have been conducted to ensure that the data in both Excel and PKL formats matched the original data in the Oracle database. This step included random sampling and cross-validation of data entries.

Accessibility and Application: By employing multiple formats, we aimed to enhance the accessibility of the dataset for a diverse range of users, including those who may prefer working within Excel or Python environments. This flexibility allows for broader applicability of the dataset in various analytical contexts and promotes collaborative research efforts.

7. Applications

1. **Identifying Key Sentiments:** This dataset can be utilized to analyze sentiments related to specific areas of technical education, such as curriculum effectiveness, student engagement, faculty support, and infrastructure. By examining the sentiments expressed in feedback, researchers can identify strengths and weaknesses in these areas.
2. **Improving Educational Practices:** Insights drawn from sentiment analysis can inform educational practices and policies. For instance, positive sentiments related to particular teaching methodologies or resources can be highlighted, while negative sentiments can prompt investigations into areas requiring improvement.
3. **Addressing Stakeholder Concerns:** This dataset allows researchers to gauge the sentiments of institutional representatives, which may reflect broader stakeholder concerns, including those of students, faculty, and industry partners. By analyzing these sentiments, educational institutions can make informed decisions that align with the needs and expectations of various stakeholders.
4. **Comparative Studies:** This dataset can also serve as a foundation for comparative studies between different institutions or regions. By analyzing sentiment trends across various technical institutions, researchers can identify best practices and areas that need attention.
5. **Future Research Directions:** This publicly available dataset encourage further research that explores specific research questions within technical education, such as:

5.1 How do sentiments vary across different categories of technical programs?

5.2 What factors contribute to positive or negative sentiments among stakeholders in technical education?

5.3 How can sentiment analysis outcomes inform policy decisions in technical education?

In summary, this dataset provides a valuable tool for advancing research in sentiment analysis specific to technical education, allowing for a deeper understanding of stakeholders' sentiments and facilitating evidence-based improvements in educational practices.

Limitations

This dataset exclusively comprises textual data, devoid of any inclusion of images, photos, audio, or video files. Consequently, its applicability is restricted to single-domain applications, focusing solely on textual content. The dataset is specifically designed and well-suited for tasks and analyses centered around text-based applications [6].

Comparison

Although there are few pre-existing datasets specifically for fine-grained sentiment analysis in technical education, datasets from related domains, such as the SemEval datasets [7] for social media and product reviews, have been compared, and the comparison is provided below (Table 4).

Table 4
Comparison between SemEval datasets and ACSS datasets.

Criteria	SemEval Datasets	ACSS Dataset (AICTE)
Data Type	Text-based (e.g., tweets, reviews)	Text-based (e.g., feedback, support requests)
Sentiment Labels	Binary to fine-grained (e.g., very positive, negative)	Likely includes sentiment labels, feedback types (appreciation, complaint, Enquiry, Grievance, support, suggestion)
Domain Scope	Varied (social media, product reviews, etc.)	Technical education and institutional support
Applications	Broad, covering multiple tasks and domains	Specific to technical education institutions
Benchmarking	Sentiment analysis, text classification, aspect-based analysis	Analyzing feedback and support requests in education
Accessibility	Widely used for benchmarking NLP models	Focused on improving technical education support systems and NLP models
	Available through SemEval challenges and workshops	Available on Harvard Dataverse

Ethics Statement

The authors have read and follow the ethical requirements for publication in Data in Brief and confirming that the current work does not involve human subjects, animal experiments, or any data collected from social media platforms.

Data Availability

ACSS (AICTE Centralized Support System) Data Set (Original data) (Harvard Dataverse)

CRedit Author Statement

Manoj Singh: Conceptualization, Methodology, Software, Data curation, Writing – original draft, Visualization, Investigation; **Subhash Panwar:** Supervision; **Sanju Choudhary:** Writing – review & editing.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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